

CEBROTONIN

(Piracetam)

NAME OF THE MEDICINAL PRODUCT
CEBROTONIN 800 mg film-coated tablets

QUALITATIVE AND QUANTITATIVE COMPOSITION
Each film-coated tablet contains 800 mg of piracetam.

Excipients
Povidone, colloidal anhydrous silica, magnesium stearate, talc, hypromellose, propylene glycol, macrogol 6000, titanium dioxide, iron oxide yellow.

PHARMACEUTICAL FORM
Film-coated tablets
Appearance: Slightly yellow to beige, smooth to shine, oblong coated tablets with breaking notch

CLINICAL INFORMATION
Indications
Treatment of the elderly with some degree of cerebral functional impairment such as loss of memory, a lack of concentration or alertness and vertigo. CEBROTONIN is indicated for patients suffering from myoclonus of cortical origin, irrespective of aetiology and should be used in combination with other anti-myoclonic therapies.

Dosage and Administration
Piracetam may be taken with or without food. The film-coated tablets should be swallowed with liquid. It is recommended to take the daily dose in two to four sub-doses.

Route of Administration
For oral use.

Adults
The daily dosage should begin at 7.2 g, increasing by 4.8 g every three or four days up to a maximum of 24 g, in two or three divided doses. Treatment with other anti-myoclonic medicinal products should be maintained at the same dosage. Depending on the clinical benefit obtained, the dosage of other such medicinal products should be reduced, if possible. Once started the treatment with piracetam should be continued for as long as the original cerebral disease persists. In patients with an acute episode, spontaneous evolution may occur over time and an attempt should be made every 6 months to decrease or discontinue the medicinal treatment. This should be done by reducing the dose of piracetam by 1.2 g every two days (every three or four days in the case of a Lance- Adams syndrome, in order to prevent the possibility of sudden relapse or withdrawal seizures).

Elderly
Adjustment of the dose is recommended in elderly patients with compromised renal function (see Precautions; Renal impairment as follows).
For long-term treatment in the elderly a regular evaluation of the creatinine clearance is required to allow dosage adaptation if needed.

Renal impairment
The daily dose must be individualized according to renal function. Refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient's creatinine clearance (Clcr) in ml/min is needed. The Clcr in ml/min may be estimated from serum creatinine (mg/dl) determination using the following formula:

$$\text{Clcr} = \frac{[140 - \text{age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dl)}} \quad (\times 0.85 \text{ for women})$$

Group	Creatinine Clearance (ml/min)	Posology and frequency
Normal	> 80	Usual daily dose, 2 to 4 divided doses
Mild	50 - 79	2/3 usual daily dose, 2 or 3 divided doses
Moderate	30 – 49	1/3 usual daily dose, 2 divided doses
Severe	20 - 29	1/6 usual daily dose, 1 single intake
	< 20	contraindicated
End-stage renal disease	-	contraindicated

Hepatic impairment
No dose adjustment is needed in patients with solely hepatic impairment. In patients with hepatic impairment and renal impairment, adjustment of dose is recommended (see dose adjustment in Renal Impairment previously).

Contraindications
This product must not be applied in case of known hypersensitivity to Piracetam or to other pyrrolidone derivatives or to any other of the ingredients, listed under "Excipients".

The product must not be applied in case of cerebral hemorrhage (e.g. hemorrhagic insult).

This product must not be applied by patients suffering from renal insufficiency at final stage.

This product must not be applied by patients suffering from Huntington's Chorea.

Warnings and precautions

Effects on platelet aggregation

Due the effect of piracetam on platelet aggregation, caution is recommended in patients with severe haemorrhage, patients at risk of bleeding such as gastrointestinal ulcer, patients with underlying disorders of haemostasis, patients with a history of haemorrhagic CVA, patients undergoing major surgery including dental surgery and patients using anticoagulants or platelet antiaggregant drugs including low dose acetylsalicylic acid.

Renal insufficiency

Piracetam is eliminated via the kidneys and care should thus be taken in cases of renal insufficiency (see Dosage & Administration).

Elderly

For long-term treatment in the elderly a regular evaluation of the creatinine clearance is required to allow dosage adaptation if needed (see Dosage & Administration).

Discontinuation

Abrupt discontinuation of treatment should be avoided as this may induce myoclonic or generalized seizures in some myoclonic patients.

Interaction

Pharmacokinetic interactions

The drug interaction potential resulting in changes of piracetam pharmacokinetics is expected to be low because approximately 90% of the dose of piracetam is excreted in the urine as unchanged drug.

Piracetam does not inhibit the human liver cytochrome P450 isoforms CYP 1A2, 2B6, 2C8, 2C9, 2C19, 2D6, 2E1 and 4A9/11 at concentrations of 142, 426 and 1422 µg/ml. At 1422 µg/ml, minor inhibitory effects on CYP 2A6 and 3A4/5 were observed. However, the K_i values for inhibition of these two CYP isoforms are likely to be well in excess of 1422 µg/ml. Therefore, metabolic interaction of piracetam with other drugs is unlikely.

Thyroid hormones

Confusion, irritability and sleep disorder have been reported during concomitant treatment with thyroid extract (T3 + T4).

Acenocoumarol

Piracetam 9.6 g/d did not modify the doses of acenocoumarol necessary to reach INR 2.5 to 3.5, but compared with the effects of acenocoumarol alone, the addition of piracetam 9.6 g/d significantly decreased platelet aggregation, β-thromboglobulin release, levels of fibrinogen and von Willebrand's factors (VIII: C; VIII: vW: Ag; VIII: vW: RCo) and whole blood and plasma viscosity.

Antiepileptic drugs

A 20 g daily dose of piracetam over 4 weeks did not modify the peak and trough serum levels of antiepileptic drugs (carbamazepine, phenytoin, phenobarbitone, valproate) in epileptic patients who were receiving stable doses.

Alcohol

Concomitant administration of alcohol had no effect on piracetam serum levels and alcohol levels were not modified by a 1.6 g oral dose of piracetam.

Pregnancy and Lactation

Pregnancy

There is no sufficient experience concerning application of Piracetam during pregnancy. Animal experimental studies do not show any direct or indirect damaging effect on pregnancy, embryo/fetal development, birth or postnatal development. Piracetam passes the placenta. Approx. 70 - 90 % of maternal piracetam plasma concentration was found in the newborn blood concentration. During pregnancy, Piracetam should only be applied if it is absolutely necessary.

Lactation

Piracetam passes over to the breast milk and should therefore not be applied during lactation period. If piracetam needs to be taken during lactation period breast-feeding should be interrupted. The decision whether breastfeeding or piracetam treatment should be interrupted, has to be done by weighing out the risks and benefits for the mother and the child.

Ability to perform tasks that require judgement, motor or cognitive skills

In view of the undesirable side effects, which were observed after the administration of the preparation, there is the possibility of influence on the ability to drive and to operate machinery and this should be taken into consideration.

Adverse Reactions

Blood and lymphatic system disorders:

Not known: haemorrhagic disorder.

Immune system disorders:

Not known: anaphylactoid reaction, hypersensitivity.

Psychiatric disorders:

Common: nervousness.

Uncommon: depression.

Not known: agitation, anxiety, confusion, hallucination.

Nervous system disorders:

Common: hyperkinesia.

Uncommon: somnolence.

Not known: ataxia, balance disorder, epilepsy aggravated, headache, insomnia.

Ear and labyrinth disorders:

Not known: vertigo.

Gastrointestinal disorders:

Not known: abdominal pain, abdominal pain upper, diarrhoea, nausea, vomiting.

Skin and subcutaneous tissue disorders:

Not known: angioneurotic oedema, dermatitis, pruritus, urticaria.

General disorders and administration site conditions:

Uncommon: asthenia.

Investigations:

Common: weight increased.

Overdosage

Symptoms and signs

No additional adverse events specifically related to overdose have been reported with piracetam. The highest reported overdose with piracetam was oral intake of 75 g wherein bloody diarrhoea with abdominal pain was most probably related to the extreme high dose of sorbitol contained in the used formulation.

Treatment

In acute, significant overdosage the stomach may be emptied by induction of emesis. There is no specific antidote for overdose with piracetam. Treatment for an overdose should be symptomatic and may include haemodialysis. The extraction efficiency of the dialyser is 50 to 60% for piracetam.

Further management should be treated as clinically indicated or as recommended by the national poisons centre, where available.

CLINICAL PHARMACOLOGY

Pharmacodynamics

Piracetam is a "nootrope" that means it is a psychotropic agent which acts directly on the brain to improve efficacy of the telencephalon in both normal subjects and those suffering from some functional deficit. This area of the brain is involved in cognition and plays a role in learning and memory, in alertness and in consciousness. Piracetam does not produce either sedation or stimulation.

Piracetam can act on the Central Nervous System in a variety of ways. It will modify neurotransmission within the brain and can help to improve the metabolic environment essential for good neuronal function. It is also a haemorrhological agent and can improve microcirculation without producing vasodilation.

When given as acute or long-term treatment for patients suffering from a functional CNS deficit, it will heighten alertness and increase cognitive function. These changes are seen as a significant increase in the alpha and beta activity with a reduction in delta activity on an EEG trace.

Piracetam will protect and restore cognitive functional capacity after cerebral trauma such as hypoxia or intoxication and after electroshock therapy.

Piracetam may be given alone or together with other drugs when treating cortical myoclonia. CEBROTONIN will reduce the duration of vestibular nystagmus.

Piracetam will improve regional oxygen and glucose uptake in the brain in patients suffering from dementia subsequent to multiple infarcts or in those with cerebral ischaemia.

Piracetam will inhibit the increased aggregation of activated platelets and - in conditions where there is abnormal rigidity of the red blood cell - it can restore deformability and the ability to pass through the microvasculature.

Pharmacokinetics

The pharmacokinetic profile of piracetam is linear and time-independent with low variability over a large range of doses among the patients. This is consistent with the high permeability, high solubility and minimal metabolism of piracetam. Plasma half-life of piracetam is 5 hours. It is increased in the elderly (primarily due to impaired renal clearance) and in patients with renal impairment. Steady state plasma concentrations are achieved within 3 days of dosing.

Absorption:

Piracetam is rapidly and extensively absorbed following oral administration. In fasted patients the peak plasma concentrations are achieved 1 hour after dosing. The absolute bioavailability of piracetam oral formulations is close to 100%. Food does not affect the extent of absorption of piracetam, but it decreases C_{max} by 17% and increases T_{max} from 1 to 1.5 hours. Peak concentrations are typically 84 µg/ml and 115 µg/ml following a single oral dose of 3.2 g and repeat dose of 3.2 g twice daily, respectively.

Distribution:

Piracetam is not bound to plasma proteins and its volume of distribution is approximately 0.6 l/kg. Piracetam crosses the blood brain barrier as it has been measured in cerebrospinal fluid following intravenous administration. In cerebrospinal fluid, the T_{max} was achieved about 5 hours post-dose and the half-life was about 8.5 hours. In animals piracetam highest concentrations in the brain were in the cerebral cortex (frontal, parietal and occipital lobes), in the cerebellar cortex and in the basal ganglia. Piracetam diffuses to all tissues except adipose tissues, crosses placental barrier and penetrates the membranes of isolated red blood cells.

Metabolism:

Piracetam is not known to be metabolized in the human body. This lack of metabolism is supported by the lengthy plasma half-life in anuric patients and the high recovery of parent compound in urine.

Elimination:

The plasma half-life of piracetam in adults is about 5 hours following either intravenous or oral administration. The apparent total body clearance is 80-90 ml/min. The major route of excretion is via urine accounting for 80 to 100% of the dose. Piracetam is excreted by glomerular filtration.

Linearity:

The pharmacokinetics of piracetam are linear over the dose range of 0.8 to 12 g.

Pharmacokinetic variables like half-life and clearance are not changed with respect to the dose and the duration of treatment.

Special patient populations

Children

No formal pharmacokinetic study has been conducted in children.

Elderly

In the elderly the half-life of piracetam is increased and the increase is related to the decrease in renal function in this population (see Section Dosage and Administration).

Renal impairment

Piracetam clearance is correlated to creatinine clearance. It is therefore recommended to adjust the daily dose of piracetam based on creatinine clearance in patients with renal impairment (see Dosage & Administration). In anuric End Stage Renal Disease patients the half-life of piracetam is increased up to 59 hours. The fractional removal of piracetam was 50 to 60% during a typical 4-hour dialysis session.

Hepatic impairment

The influence of hepatic impairment on the pharmacokinetics of piracetam has not been evaluated. Because 80 to 100% of the dose is excreted in the urine as unchanged drug, hepatic impairment solely would not be expected to have a significant effect on piracetam elimination.

Other patient characteristics

Gender

At a dose of 2.4 g, C_{max} and AUC were approximately higher in women compared to men. However, clearances adjusted for body weight were comparable.

Race

Piracetam is primarily renally excreted and there are no important racial differences in creatinine clearance, pharmacokinetic differences due to race are not expected.

Non-Clinical Information:

Piracetam has a low toxicity potential. An i.v. administration of up to 1 g/kg/day for 4-5 weeks in rats and dogs did not produce toxicity. Piracetam has shown no potential for genotoxicity and carcinogenicity.

PHARMACEUTICAL INFORMATION

Shelf life

The product has a shelf life of 3 years.

The product should not be applied any more after its expiry.

Shelf life after opening the package or after production of the ready for use preparation: Not applicable.

Storage

To be stored in a dry place below 30°C, protected from light and out of the reach of children.

Nature and contents of container

Pack size

Box of 30 film-coated tablets. 3 blister strips per box, each strip containing 10 film-coated tablets.

Incompatibilities

Not known so far.

Special precautions for disposal and other handling

None.

Manufactured by

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